

Modeling Generalization via Out-of-Distribution Detection under Covariate Shift in Visual Deep Neural Networks

Anonymous CVPR submission

Paper ID *****

Abstract

Deep learning vision systems are known to be sensitive to data that exhibit certain properties not well represented in the training data. When data of this nature manifest in real-world settings, it is often referred to as a covariate shift and is associated with a performance gap compared to initial evaluations. In this paper, we study the extent to which these types of performance gaps and the shifts that precede them can be effectively predicted using out-of-distribution (OOD) detection algorithms. OOD detectors have been studied extensively in the context of open-set recognition and semantic shift detection, often achieving high accuracy. However, covariate shifts represent a more difficult detection target due to the high similarity of the resulting data to in-distribution data, as well as the varying scope and severity that may characterize these shifts. We therefore assess the circumstances and conditions under which OOD detectors can successfully detect covariate shifts. We observe that there is a strong relationship between the rate at which OOD detectors return positive verdicts for a given shift and the generalization gap that is typically induced. We exploit this relationship to build predictive models of generalization, which we demonstrate can provide low-error estimates of network accuracy, with the best-performing OOD detectors attaining mean absolute errors ranging from 0.04 to 0.13, out-performing competing methods. We position OOD detectors as underutilized tools for providing deep neural networks with a greater degree of reliability under covariate shift.

1. Introduction

Despite their remarkable performance on benchmarks compared to competing approaches, visual Deep Neural Networks (DNNs) has been shown to frequently fail in real-world scenarios, a phenomenon known as *generalization failure* [3, 14, 15, 21, 21, 38, 64]. In essence, these failures arise from a misalignment between the nature of the data

on which DNNs is typically trained and evaluated and the broader, more variable data distributions encountered in deployment. This misalignment is commonly described as a *covariate shift* and adversely impacts DNN reliability. Under covariate shifts, DNNs are not guaranteed to yield correct predictions at the rate specified from initial evaluations, which are typically based on In-Distribution (InD) data extracted from the overall training set. These limitations pose serious challenges to the safe deployment of visual DNNs, particularly in high-stakes domains such as medical imaging and autonomous vehicles.

Although significant strides have been made with respect to improving the generalization capabilities of DNNs under covariate shift [51], notably through domain-adaptation frameworks [19, 62] and, recently incorporating text-corpora [31], attaining sufficient levels of generalization for responsible deployment in high-risk settings remains an open problem. As a result, emerging work is instead focusing on the implementation of support components that can mitigate the risks associated with poor generalization. Notable examples thereof include the implementation of guardrails for Large Language Models (LLMs) [4, 69] and abstention classes [30, 54]. *Out-of-Distribution (OOD) detectors* have recently also been explored as a means of ensuring output reliability through the operationalization of abstention in DNNs [56, 57]. In other work, OOD detectors have been demonstrated to facilitate label-free runtime estimation of loss [58] and accuracy [55]. However, as OOD detectors have primarily been studied in the context of open-set recognition and class novelty rather than in terms of characterizing generalization, their capabilities under covariate shift are not well understood.

In this paper, we study how aggregate OOD detector verdict rates can be used to characterize covariate-shift severity and estimate sequence-level generalization performance. While prior work has shown that OOD behavior is coupled to classifier accuracy and shift structure, we operationalize this relationship into a simple label-free accuracy estimator under covariate shift. In particular, we assess the detection rates of several popular OOD detector implementations on

several types and intensities of covariate shift, across multiple datasets, and their relationship to generalization. In general, we observe that OOD detection rates under covariate shift are modulated by the size of the generalization gap between the shifted data and the InD data. We observe a fairly strong relationship between OOD detector prediction rates and generalization, albeit with varying degrees of linearity and noise. Motivated by these findings, we develop a novel approach to label-free accuracy estimation, where we train a linear model that maps the OOD detection rate for a sequence of data to an estimated generalization gap. Our approach attains absolute errors ranging between 0.04 and 0.13, outperforming competing approaches.

We summarize our contributions as follows:

1. We assess the capabilities of several OOD detector implementations with respect to detecting different types and intensities of covariate shift.
2. We demonstrate a consistent relationship between generalization gaps and OOD detector performance, showing that detection is more reliable when shifts induce substantial performance degradation. This highlights both the potential and the limitations of OOD detection as a covariate shift safeguard.
3. We exploit this relationship to build models to predict generalization on novel data and achieve mean absolute errors ranging between 0.04 and 0.13 over a wide range of different covariate shifts.

The remainder of this paper is organized as follows: Section 2 discusses adjacent and related work. Section 4 outlines the methodology used as part of our study. Section 5 then shows our results. We then discuss these results in Section 6 and conclude our work in Section 7. All code used as part of this work is available at our GitHub Repository¹.

2. Related Work

Distributional shift, also often referred to as dataset shift, is a well-known threat to the reliability of neural networks [41]. Broadly, distributional shifts can be distinguished according to the variables they affect:

- A shift in $P(y)$, i.e., the class distribution, is referred to as a *label shift* or a *prior probability shift* [41].
- A shift in $P(x)$ is referred to as a *covariate shift* and encompasses changes to the nature of the data while the relationship between the input and output variables remains the same.

- A shift in $P(y|x)$ is referred to as a *semantic shift* and encompasses changes to the data to which a DNN cannot be expected to generalize, since it represents a completely different data generation process.

Label shift is often considered relatively benign. Although a shift in the distribution of labels can indeed induce performance degradation, particularly for datasets with class imbalance [5], modern DNN are often made more resilient to these types of shifts through, e.g., importance sampling. Generalization to semantic shifts is generally infeasible, as this corresponds to data on which a given DNN simply is not designed to perform inferences. This may, for instance, be a medical imaging system encountering non-medical images, or a classifier encountering classes beyond the scope of its output space [66]. Covariate shifts, however, represent changes in the data to which an ideally trained DNN may be expected to generalize [41, 56]. DNN often fails to generalize under covariate shift; however, such shifts correspond to a violation of the Independently and Identically distributed (IID) assumption, undermining the validity of the predictor returned from the training process [59]. This is referred to as *generalization failure*. Generalization failure is prevalent in a wide variety of DNN domains, notably medical imaging [3, 8, 15, 21, 64], hybrid language vision models [31], LLMs [7, 11], and even benchmarking datasets [46, 47].

Several methods have been proposed to address this sensitivity to covariate shift [51]. Several alternatives to Empirical Risk Minimization (ERM) have been proposed, for instance [6, 20, 66], although it is contested whether these methods actually outperform ERM outside of certain narrow conditions [68]. Uncertainty quantification has also been cited as a potential solution [1], but it has been shown that the quality of the resulting uncertainty estimates often degrades in OOD settings [43]. Other works conceptualize *domain adaptation* as a plausible solution, where incoming samples are aligned to a model of the training distribution [19, 62]. While there is significant progress in this field, some works suggest that these methods are only effective for data characterized by certain specific properties [39]. Finally, the emergence of Contrastive Language-Image Pre-training (CLIP) and LLM has permitted visual models to leverage text corpora as part of training and has been shown to often increase generalization [18, 31]. However, other analyses indicate that such models are often subject to similar pitfalls in terms of shortcut learning and poor generalization as regular vision models [63].

Rather than attempting to increase model generalization, some have investigated methods by which the reliability of DNN outputs can be assured through other means. Notable examples of this include the implementation of abstention classes [54], such that DNN classifiers can defer predictions when there is insufficient basis for a reliable prediction. In LLM research, similar approaches have been pro-

¹<https://anonymous.4open.science/r/GeneralizationPredictionCovariateShift-6A51>

posed in order to reduce the incidence of hallucinations [30] or to avoid unsafe outputs [4, 28, 69]. Recent work has highlighted that OOD detectors may operationalize this approach in vision domains [56–58].

OOD detectors are intended to detect instances of data with near-zero density in the (learned) data distribution, i.e. $P(x, y) \approx 0$. There exists a wide variety of means by which this can be achieved. The first proposed methods of OOD detection were based on computing simple features from the output logits, such as maximum softmax scores or entropy scores [25]. More advanced methods, like ODIN, augment this through temperature scaling and adversarial perturbation [35]. Energy and other information-theoretic features derived from output logits are also common [36]. Gradient-based methods have also been the subject of significant research, though the mechanisms underpinning their efficacy for OOD detection are somewhat contested [26, 27, 33]. All of these methods entail computing features that are assumed to more or less approximate $P(x, y)$ in a much more tractable fashion. Classification is then performed by setting a threshold according to known InD (and OOD) data [66]. More direct density-based methods have also been studied, notably in the form of statistical testing [10, 42, 44].

OOD detectors have, in recent work, been leveraged to build models for the label-free prediction of the loss [58] and accuracy [55] associated with a set of predictions. These methods are respectively based on building regression models on OOD detector features or by computing probabilistic estimates of accuracy according to the incidence rate of distributional shift as estimated from OOD verdict traces.

However, as OOD detectors have primarily been studied in the context of semantic shift [45, 66], it is hitherto underexplored whether OOD detectors can reliably detect covariate shifts of differing character and intensity. Notably, it was observed as part of the analysis of the aforementioned accuracy estimation method that the estimates are contingent on the assumption that all distributional shifts are associated with an approximately similar loss of accuracy. As was shown in the work on loss-regression, however, different intensities of shift give rise to proportionally higher loss.

3. OOD Detection and Covariate Shifts

DNNs in vision tasks are generally trained in a supervised fashion through ERM. ERM seeks to solve the optimization problem $\hat{h} = \arg \min_h \int l(h(x), y) dP(x, y)$, i.e., to find the network parameter configuration $\hat{h}$ that is characterized by minimum loss $l(\cdot)$ over the distribution of input-output data $P(x, y)$. As this distribution is intractable, it is assumed that it can be approximated empirically by a dataset $\mathcal{D} = \{x_i, y_i\}_0^n$ through IID sampling. The ERM objective is thus given by $R(h) = \sum_{i=0}^n l(h(x_i), y_i)$.

The IID assumption is frequently violated in practice, as datasets are often characterized by some degree of latent bias [15, 21]. This greatly impedes the generalizability of the resulting DNN, as it is unlikely that any given network will infer the correct predictive patterns for inputs that lack support in the training data.

This notion can be formalized as follows: let $P(x, y)$ be the total data distribution that the dataset $\mathcal{D}$ seeks to model empirically. When the IID violation is violated, the dataset can be considered to follow a different distribution $\mathcal{D} \sim P_{train}(x, y)$, $P_{train}(x, y) \neq P(x, y)$. In this case, the network training will not be significantly influenced by data where $P_{train}(x, y) \approx 0$ and $P(x, y) \gg 0$, potentially causing a generalization gap.

OOD detectors are intended to detect instances of data with low density in the distribution $P(x, y)$, with OOD data commonly defined by the following relation [66]:

$$P(x, y) \approx 0 \quad (1)$$

Distributional shifts, on the other hand, are defined by the relation:

$$P_{train}(x, y) \neq P_{test}(x, y) \quad (2)$$

While the two are not typically considered simultaneously in research, OOD detection can largely be regarded as a means of distributional shift detection. A distributional shift effectively corresponds to a change in the distribution such that samples drawn from the new distribution can, depending on the severity of the shift, have a near-zero probability in the original training distribution. Thus, by detecting when incoming data has a low probability in P_{train} , it is therefore possible to classify data instances according to whether a distributional shift has occurred. Depending on the implementation, this may encompass all three different types of distributional shift listed previously.

Recent work has increasingly questioned whether standard OOD detectors primarily measure semantic novelty, covariate shift, classifier quality, or some mixture thereof. Vaze et al. [60] show that open-set recognition performance is strongly coupled to closed-set classification accuracy, while Miller et al. [40] demonstrate strong correlations between in-distribution and out-of-distribution generalization across several benchmarks. Rody et al. [48] similarly show that open-set/OOD detection becomes more difficult as out-of-distribution data becomes more similar to in-distribution data. More recent work has further problematized standard OOD evaluation: Yang et al. [67] argue that modern OOD detectors are often highly sensitive to covariate shifts; Long et al. [37] propose a fine-grained benchmark spanning semantic and covariate shift degrees; and Li et al. [34] argue that many supervised OOD detectors answer questions misaligned with semantic novelty detection. Harun et al. [23] provide a representation-geometric account connecting OOD behavior and generalization.

Our work is complementary to these findings. Rather than treating this coupling as a failure mode for semantic novelty detection, we study whether detector verdict rates can be used constructively as sequence-level covariate-shift statistics for label-free accuracy estimation.

4. Methodology

In this section, we outline the methodology used for our analysis. Broadly, we seek to characterize the degree to which different types of OOD detectors are capable of detecting instances of covariate shift of various types and severities across several datasets. To this end, we consider 5 different datasets and 6 different post-hoc OOD detector implementations. We also apply varying intensities of synthetic covariate shifts in addition to the organic covariate shifts already present in the datasets.

4.1. Datasets

To sufficiently study the effects of covariate shifts, we require datasets that contain organic instances of such shifts. To this end, we utilize several domain adaptation datasets, as well as a polyp segmentation benchmark consisting of four independently curated datasets. An overview of these datasets is provided in Table 1.

4.2. Synthetic shifts

As the aforementioned datasets contain only one type of covariate shift, we also utilize synthetic shifts to assess how OOD detectors respond to different types and intensities of covariate shifts. To this end we bootstrap new datasets from the InD validation dataset and apply a given intensity and type of data augmentation. The intensity is determined by a parameter α , chosen by visual inspection, such that the classes could still be readily distinguished at maximum intensity. The augmentations are listed in Table 2.

4.3. OOD Detectors

We study a selection of OOD detectors based on the taxonomy provided in Reference [66]. We narrow the scope to post-hoc methods. An overview of the studied OOD detectors is provided in Table 3. We calibrate the OOD detectors by finding the optimal threshold between the InD data and one organic OOD partition. We then evaluate the OOD detector accuracies on the remaining partitions. This is to avoid any biases from setting the thresholds manually and represents a realistic choice in engineering settings.

4.4. Network training

For the purposes of generality, we perform assessments on several different architectures. We train a ResNet101 [24], a vision-transformer [16] for all the classification datasets, and a DeepLabV3, UNet [49], and Segformer [65]

for the polyp segmentation datasets. All models are pre-trained on ImageNet. The training configuration is summarized in Table 4.

4.5. Evaluation

As the labeling of covariate shifts is ambiguous, we do not assess the OOD detectors in terms of classification accuracy. Instead, we assess the detection rate observed over different covariate shifts, and how they relate to the generalization gap induced by said shift. We expect that shifts of greater intensity as given by the α parameter induce a correspondingly greater generalization gap, and that they are more readily detected as OOD.

5. Results

In this section, we first evaluate the extent to which OOD detector verdicts are predictive of generalization. We observe that there is frequently a fairly strong, albeit imperfect, positive relationship between OOD detection rates and generalization gaps. We then exploit this relationship to build predictive models of generalization, which attain lower errors compared to competing approaches.

5.1. Shift Detectability and Generalization

Covariate shifts differ from semantic shifts in that they can manifest in different degrees of performance degradation. We therefore assess the frequency at which different covariate shifts, with differing induced generalization gaps, are detected. This was achieved by applying varying intensities of synthetic shifts to the InD validation fold and computing the resulting accuracy of the DNN. For brevity, we present the combination of OOD detector and network architecture with relationships that are characterized by the strongest correlations for each dataset. The results in Figure 1 show a fairly strong positive relationship between the OOD detection rate for a given shift and the generalization gap it induces. Although we do not expect these relationships to persist uniformly across all detector features and architectures, their emergence nonetheless reveals that DNNs may have a latent capacity for characterizing covariate shifts given an appropriate design.

This property can be understood in terms of the density of the data in the learned distribution. Data that has undergone a greater degree of covariate shift has a lower density in the learned distribution. OOD detectors approximate this density through various heuristics, albeit with varying degrees of fidelity. As a result, OOD detection rates represent an approximate signal of density. As lower density data has comparatively smaller effects on training, it follows that it is less likely to be associated with correct predictions, discounting the influence of any inductive biases that may otherwise contribute to accurate predictions. The well-behavedness of the relationship is therefore dependent

Table 1. Overview of Datasets.

Dataset	No. Images	Resolution	No. Classes	Train/Test Domains	Description
NICO [70]	230000	Varying	60	1/5	domain-adaptation classification dataset partitioned by a number of contexts.
Office-Home [61]	15500	Varying	65	1/3	A domain-adaptation classification dataset partitioned by modality
Office-31 [50]	4110	Varying	31	1/2	A domain-adaptation classification dataset partitioned by camera type
CCT [9]	243187	1024X747	15	65/65	a domain-adaptation/generalization benchmark consisting of images taken from different camera-traps.
Polyps [2, 12, 29, 52]	2720	arying	1	1/3	Polyp segmentation datasets collected from different centres. We use Kvasir as the InD dataset.

Table 2. Overview of synthetic augmentations.

Shift	Functional Description	α Range	Real-world analogue
Additive noise	$x = x + \alpha U(-1, 1)$	$(0, 0.5]$	Image noise, interference, etc.
Multiplicative Noise	$x = x + \alpha U(-1, 1)x$	$(0, 0.5]$	Image noise, interference, etc.
Salt and Pepper Noise	$\begin{cases} x[U(-1, 1) < \alpha] = 0 \\ x[U(-1, 1) > \alpha] = 1 \end{cases}$	$(0, 0.125]$	Image noise, interference, faulty camera sensor, etc.
FGSM [13]	$x = x + \alpha \text{sign}(\nabla_x(L(\theta, x, y)))$	$[0, 0.5]$	Adversarial attacks
Hue	HueJitter(α)	$[0.05, 0.3]$	Poor camera color correction, lighting changes
Brightness	BrightnessJitter(α)	$[0.05, 1]$	Over- and under-exposure, changes to lighting conditions
Contrast	ContrastJitter(α)	$(0, 1]$	Changes to camera settings/hardware
Smear	GridDistortion(α)	$(0, 0.5]$	Camera movement
JPEG-compression	ImageCompression(α)	$(0, 75)$	Image Quality
Fog	RandomFog(α)	$(0, 0.5]$	Fog/Partial Occlusion

on the fidelity with which OOD detectors express the density of a given datum in the learned distribution and any influence of inductive biases. There may, as a result, exist certain shifts that break this pattern, e.g., by eluding detection despite inducing significant generalization gaps. This can be observed to differing degrees

5.2. Accuracy Prediction

We exploit the relationship observed above to build predictive models of generalization. This is achieved by fitting a linear model to the data. While the observed relationships are somewhat non-linear, we impose this linear constraint to facilitate accurate predictions on sequences composed of different types and intensities of shifts. Since the overall de-

tection rate and accuracy of a given sequence is a weighted sum of the detection rates and accuracies associated with the constituent sub-sequences, it follows that a predictive model of accuracy must also exhibit this linear property.

We assess the accuracy of the resulting predictions as follows: we select one shift as a hold-out set for evaluation and train the regression model on the data corresponding to the remaining shifts. We then bootstrap several sequences composed of a varying proportion of InD data and data of a particular intensity of shift. The Mean Average Error (MAE) in the accuracy estimates are then computed over these sequences. We then compute the average MAE over over shift types, intensities, and composition ratios. The results are shown in Table 5, selecting the lowest-error network archi-

Table 3. Overview of OoD detectors.

Detector	Formula	Description
Entropy [25]	$H(x) = -\sum_y p_\theta(y x) \log p_\theta(y x)$	Predictive entropy of the model output distribution.
GradNorm [26]	$\ \nabla_\theta H(x)\ _2$	l_2 norm of the gradient of entropy w.r.t. model parameters.
Energy [36]	$E(x) = -\log \sum_y e^{f_\theta(x)_y}$	Energy score, i.e. negative log-sum-exp of logits $f_\theta(x)$.
kNN [53]	$\min_{x_i \in \mathcal{D}_{\text{train}}} \ z(x) - z(x_i)\ _2$	Distance between test encoding $z(x)$ and its nearest neighbour $z_k(x)$ in the training set. The original paper extends this to multiple k values.
MSP [25]	$\max(p_\theta(y x))$	Maximum softmax score.
Typicality [42]	$-\log p_{\text{GLOW}}(x)$	Negative log-likelihood under the GLoW generative model.

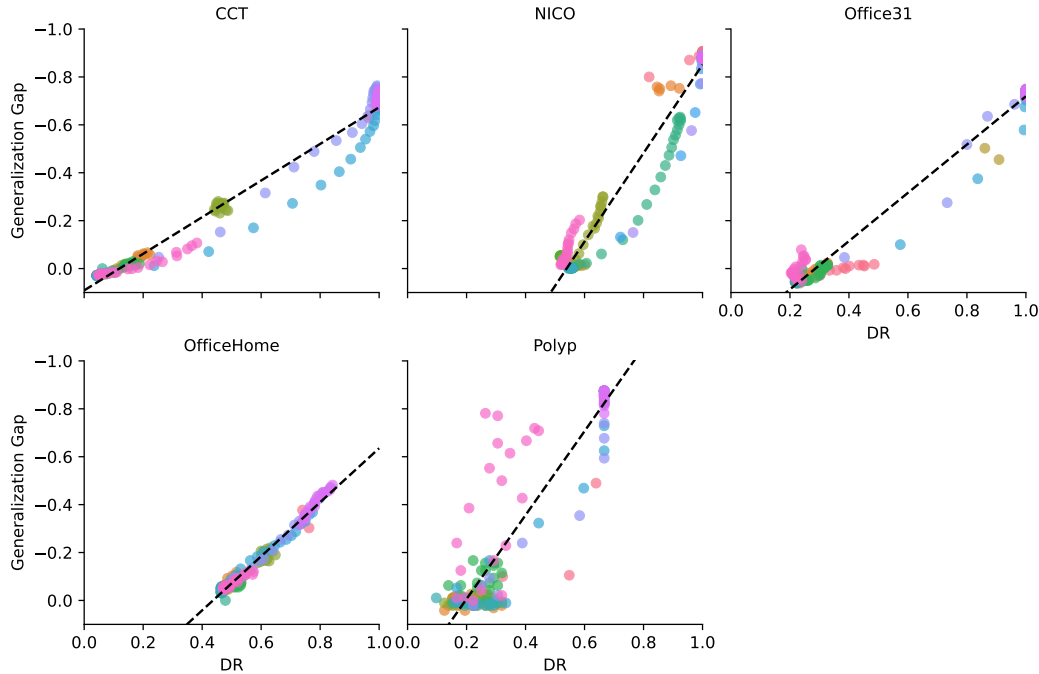


Figure 1. Strongest relationships between OOD detection rate and generalization gap across covariate shifts.

ture for each configuration. We compare these errors to a baseline of assuming InD generalization, as well as the error of the estimates provided by Probabilistic Runtime Evaluation (PRE), the closest competing approach.

The results indicate that our proposed approach, on average, provides lower error estimates of generalization compared to the baseline approach, with MAEs ranging between 0.04 and 0.13. Our approach generally outperforms PRE, with the exception of OfficeHome, for which PRE attains slightly better results.

These error rates are averaged over shift intensities and InD proportions. To provide a more granular view, we break

down these results according to the proportion of OOD data in a given sequence and the intensity of the shifts. Figure 2 shows heatmaps of the resulting error estimates. The red, blue, and green palettes respectively correspond to whether the baseline, PRE, or our approach is associated with the lowest error.

The results affirm that our approach generally outperforms the competing approaches, albeit with a few caveats. Notably, the naive baseline often yields lower errors for low-intensity shifts and in cases where data sequences are characterized by a lower proportion of InD data. This follows naturally from the fact that data sequences of this na-

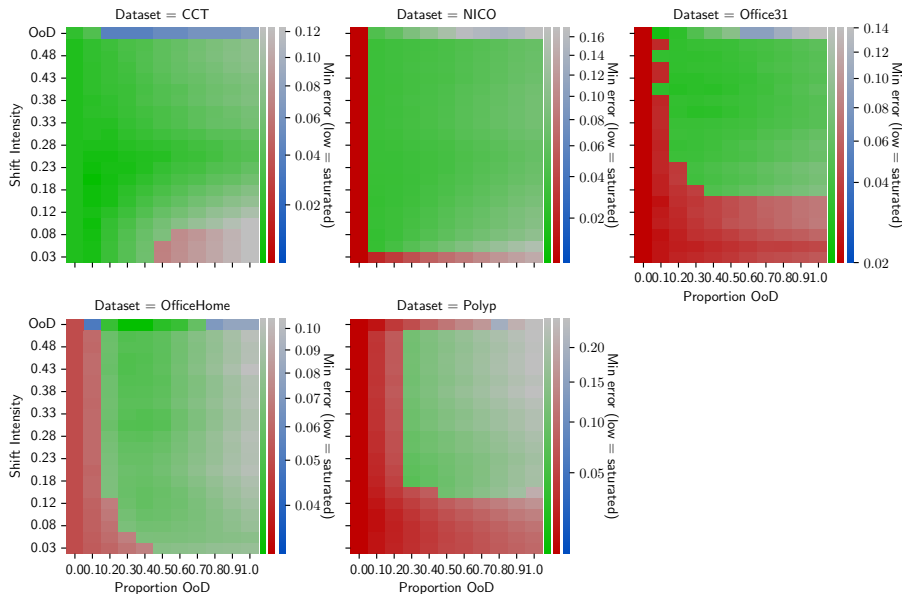


Figure 2. Comparison to baselines across shift intensities and InD proportions.

Table 4. Overview of Training Method.

Component	Choice
Framework	Pytorch Lightning [17].
Pretraining	Imagenet
Optimizer	Adam [32]
Scheduler	Cosine Annealing w/ Warm Restarts
Batch size	16
Data augmentation	Random flip, Random rotate
Initial learning Rate	1e-3
Model Selection	Early Stopping
Loss	Cross Entropy Loss / Jaccard loss

ture are characterized by near-InD performance and thus low generalization gaps. PRE, on the other hand, often outperforms our approach for the organic OOD data. This is due to the fact that PRE assumes that the generalization gap of the calibration dataset is applicable to all types of shifts, which does not hold for shifts of significantly differing character.

In practice, accuracy estimation may provide an increased degree of transparency with respect to the reliability of network outputs under operational conditions. However, if our approach provides high-error estimates for InD data, any value judgments will be excessively conservative. We therefore assess how the errors vary across the generalization gaps associated with each tested shift and intensity in

Figure 3.

The results show that the estimation errors are, in general, weakly correlated with generalization. Notably, however, there is some degree of variability in the errors, spanning orders of magnitude. On average, as indicated by the red sliding window average, our approach generally outperforms the baseline for data with low generalization, albeit by differing margins depending on the dataset. However, For shifts associated with low generalization gaps, the errors sometimes exceed the baseline.

6. Discussion

Overall, our results indicate that OOD detectors are fairly effective as covariate shift detectors, insofar as they are relatively expressive of generalization gaps on sequences of data. We frequently observed high OOD detection rates for high-intensity shifts, highlighting their ability for runtime verification. For low intensity shifts, the detection rates were correspondingly lower, but in a predictable manner that coincided with the induced generalization gap. We leveraged this property to build predictive models of generalization, attaining fairly low errors and outperforming the baselines. There are a few caveats to these results, however, which we discuss further in the following paragraphs.

The results should not be interpreted as evidence that OOD detectors universally solve covariate-shift detection. Rather, they support a narrower claim: when detector scores vary monotonically with shift-induced degradation, aggregate detection rates provide a useful signal for stream-level

	CCT	NICO	Office31	OfficeHome	Polyp
Energy	0.13	0.05	0.09	0.11	0.14
Entropy	0.24	0.25	0.10	0.12	0.13
GradNorm	0.27	0.06	0.26	0.10	0.35
Softmax	0.16	0.11	0.13	0.07	—
Typicality	0.05	0.26	0.05	0.21	0.25
kNN	0.04	0.15	0.11	0.09	0.17
Baseline	0.17	0.18	0.14	0.12	0.17
Best PRE [55]	0.05	0.16	0.10	0.06	0.14
Best (Ours)	0.04	0.05	0.05	0.07	0.13

Table 5. MAE comparison for each OOD detector and baseline across datasets.

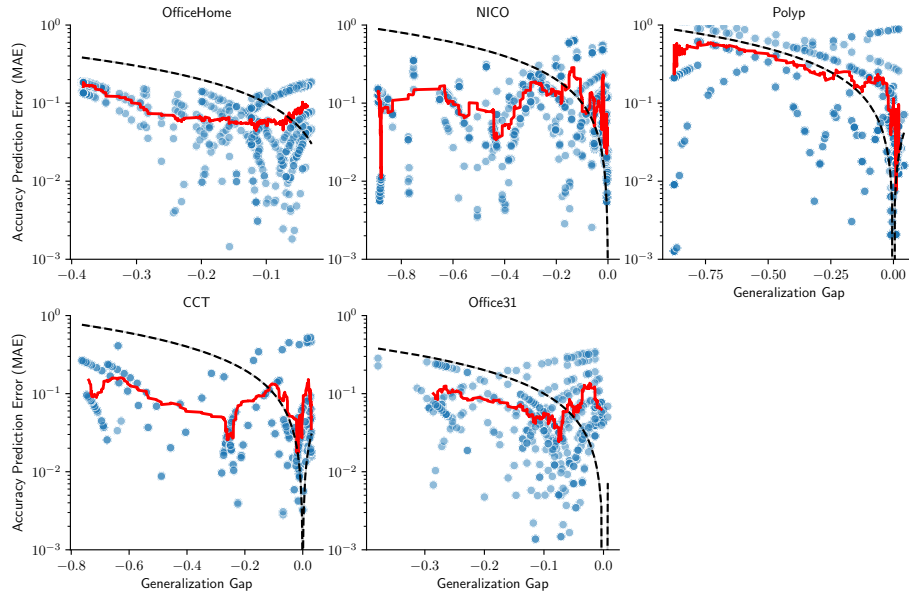


Figure 3. MAEs across generalization gaps compared to the baseline (black). The red line is a sliding window average over the errors.

reliability estimation. This interpretation is consistent with recent critiques showing that supervised OOD detectors often respond strongly to covariate rather than semantic shifts [34, 67]. In our setting, this behavior is useful, but it also implies that detector choice, thresholding, architecture, and training recipe must be validated for the deployment distribution.

We note that our experiments all consider sequences of data, rather than individual data points. However, the strength of the relationships we observe may be modulated by sequence length. Moreover, while the detection rates were observed to coincide strongly with accuracy, it may not be the case that OOD detectors reliably detect the particular individual data associated with incorrect predictions, as opposed to broadly characterizing a given shift and its influence asymptotically. While this work has demonstrated that OOD detectors have significant promise for detecting co-

variate shifts, a stronger runtime-verification-oriented perspective is warranted, e.g., through a study of the precise capabilities of OOD detectors for classifying the correctness of individual predictions. We assert that the results presented in this work represent compelling initial evidence for the efficacy of OOD detectors as such.

However, it is worth noting that the well-behavedness of the relationships observed in this work may not necessarily generalize. Different OOD detector algorithms and model architectures may exhibit weaker or stronger relationships than what we observed in this work. Moreover, certain shifts may yield somewhat inconsistent behavior, as can be observed in Figure 1. In spite of this, the mere existence of these relationships point to an underexplored latent capability in DNNs for covariate shift characterization and management. With continued research on more OOD detection features, it may be possible to construct methods with

the explicit intent of attaining high correlations between detection rates and generalization.

We also recognize that a more systematic analysis is warranted. We have considered a fairly small selection of OOD detectors in our experiments. An understanding of precisely why different types of OOD detectors succeed or fail to detect certain types of shifts would be of significant merit, but requires a larger number of different implementations, datasets, and synthetic shifts.

We have primarily considered fairly simple synthetic shifts in the form of data augmentations in this work. However, more sophisticated augmentation strategies are available. Future analyses may, for instance, benefit from assessment under synthetically generated data from generative models or model-based augmentation [22]. Beyond this, it would be useful to curate datasets with different intensities of organic shifts where possible and, in general, to develop a more extensive research infrastructure for the study of covariate shifts.

A further limitation is that our models use relatively standard fine-tuning recipes. Stronger robustness-oriented training procedures, including RandAugment, MixUp, CutMix, random resized cropping, and self-supervised pre-training, may alter both the magnitude of the generalization gaps and the monotonicity of the detector-rate relationship. Evaluating whether the proposed estimator remains calibrated under such recipes is an important direction for future work.

7. Conclusion

In this paper, we have studied the capabilities of OOD detectors as covariate shift detectors. We observed that OOD detection rates tend to increase fairly monotonically with larger generalization gaps, which we exploited to build predictive models of generalization that outperformed baseline approaches. This suggests that OOD detectors may operate as simple runtime verifiers under covariate shift and therefore provide functionality for classifying whether the prediction of a given datum should be deferred. We invite further work for assessing the capabilities of OOD detectors under covariate shift and for the purposes of reliability enhancement through runtime performance characterizations and runtime verification.

References

- [1] Moloud Abdar, Farhad Pourpanah, Sadiq Hussain, Dana Rezazadegan, Li Liu, Mohammad Ghavamzadeh, Paul Fieguth, Xiaochun Cao, Abbas Khosravi, U. Rajendra Acharya, Vladimir Makarenkov, and Saeid Nahavandi. A review of uncertainty quantification in deep learning: Techniques, applications and challenges. *Information Fusion*, 76:243–297, 2021. 2
- [2] Sharib Ali, Christian Daul, Jens Rittscher, Danail Stoyanov, and Enrico Grisan, editors. *EndoCV2020: 2nd International Workshop and Challenge on Computer Vision in Endoscopy*, volume 2595, Iowa, USA, 2020. CEUR Workshop Proceedings. 5
- [3] Sharib Ali, Noha Ghatwary, Debesh Jha, Ece Isik-Polat, Gorkem Polat, Chen Yang, Wuyang Li, Adrian Galdran, Miguel-Ángel González Ballester, Vajira Thambawita, Steven Hicks, Sahadev Poudel, Sang-Woong Lee, Ziyi Jin, Tianyuan Gan, Chenghui Yu, JiangPeng Yan, Doyeob Yeo, Hyunseok Lee, Nikhil Kumar Tomar, Mahmood Haithami, Amr Ahmed, Michael A. Riegler, Christian Daul, Pål Halvorsen, Jens Rittscher, Osama E. Salem, Dominique Lamarque, Renato Cannizzaro, Stefano Realdon, Thomas de Lange, and James E. East. Assessing generalisability of deep learning-based polyp detection and segmentation methods through a computer vision challenge. *Scientific Reports*, 14(1):2032, Jan 2024. 1, 2
- [4] Gabriel Alon and Michael Kamfonas. Detecting language model attacks with perplexity, 2023. 1, 3
- [5] R. Anand, K.G. Mehrotra, C.K. Mohan, and S. Ranka. An improved algorithm for neural network classification of imbalanced training sets. *IEEE Transactions on Neural Networks*, 4(6):962–969, 1993. 2
- [6] Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization, 2019. 2
- [7] Razvan Azamfirei, Sapna R Kudchadkar, and James Fackler. Large language models and the perils of their hallucinations. *Critical Care*, 27(1):120, 2023. 2
- [8] Emma Beede, Elizabeth Baylor, Fred Hersch, Anna Iurchenko, Lauren Wilcox, Paisan Ruamviboonsuk, and Laura M. Vardoulakis. A human-centered evaluation of a deep learning system deployed in clinics for the detection of diabetic retinopathy. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, CHI '20, page 1–12. Association for Computing Machinery, 2020. 2
- [9] Sara Beery, Grant Van Horn, and Pietro Perona. Recognition in terra incognita, September 2018. 5
- [10] Federico Bergamin, Pierre-Alexandre Mattei, Jakob Drachmann Havtorn, Hugo Sénétaire, Hugo Schmutz, Lars Maaløe, Soren Hauberg, and Jes Frellsen. Model-agnostic out-of-distribution detection using combined statistical tests. In *Proceedings of The 25th International Conference on Artificial Intelligence and Statistics*, volume 151 of *Proceedings of Machine Learning Research*, pages 10753–10776. PMLR, March 2022. 3
- [11] Lukas Berglund, Meg Tong, Max Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak, and Owain Evans. The reversal curse: LLMs trained on "a is b" fail to learn "b is a", 2023. 2
- [12] Jorge Bernal, F Javier Sánchez, Gloria Fernández-Esparrach, Debora Gil, Cristina Rodríguez, and Fernando Vilarinho. Wm-dova maps for accurate polyp highlighting in colonoscopy: Validation vs. saliency maps from physicians. *Computerized Medical Imaging and Graphics*, 43:99–111, 2015. 5
- [13] Battista Biggio, Igino Corona, Davide Maiorca, Blaine Nelson, Nedim Šrndić, Pavel Laskov, Giorgio Giacinto, and

- Fabio Roli. Evasion attacks against machine learning at test time. *Lecture Notes in Computer Science*, page 387–402, 2013. 5
- [14] Krzysztof Budzyń, Marcin Romańczyk, Diana Kitala, Paweł Kołodziej, Marek Bugajski, Hans O. Adami, Johannes Blom, Marek Buszkiewicz, Natalie Halvorsen, Cesare Hassan, Tomasz Romańczyk, Øyvind Holme, Krzysztof Jarus, Shona Fielding, Melina Kunar, Maria Pellise, Nastazja Pilonis, Michał Filip Kamiński, Mette Kalager, Michael Bretthauer, and Yuichi Mori. Endoscopist deskilling risk after exposure to artificial intelligence in colonoscopy: a multicentre, observational study. *The Lancet Gastroenterology & Hepatology*, 8 2025. 1
- [15] Alexander D’Amour, Katherine Heller, Dan Moldovan, Ben Adlam, Babak Alipanahi, Alex Beutel, Christina Chen, Jonathan Deaton, Jacob Eisenstein, Matthew D. Hoffman, Farhad Hormozdiari, Neil Houlsby, Shaobo Hou, Ghasen Jerfel, Alan Karthikesalingam, Mario Lucic, Yian Ma, Cory McLean, Diana Mincu, Akinori Mitani, Andrea Montanari, Zachary Nado, Vivek Natarajan, Christopher Nielson, Thomas F. Osborne, Rajiv Raman, Kim Ramasamy, Rory Sayres, Jessica Schrouff, Martin Seneviratne, Shannon Sequeira, Harini Suresh, Victor Veitch, Max Vladymyrov, Xuezhi Wang, Kellie Webster, Steve Yadlowsky, Taedong Yun, Xiaohua Zhai, and D. Sculley. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61, 2022. 1, 2, 3
- [16] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale, 2021. 4
- [17] William Falcon and The PyTorch Lightning team. PyTorch Lightning, Mar. 2019. 7
- [18] Alex Fang, Gabriel Ilharco, Mitchell Wortsman, Yuhao Wan, Vaishaal Shankar, Achal Dave, and Ludwig Schmidt. Data determines distributional robustness in contrastive language image pre-training (CLIP). In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 6216–6234. PMLR, 7 2022. 2
- [19] Abolfazl Farahani, Sahar Voghoei, Khaled Rasheed, and Hamid R. Arabnia. A brief review of domain adaptation. In *Advances in Data Science and Information Engineering*, pages 877–894, Cham, 2021. Springer International Publishing. 1, 2
- [20] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. 2015. 2
- [21] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, Nov. 2020. 1, 2, 3
- [22] Sven Gowal, Sylvestre-Alvise Rebuffi, Olivia Wiles, Florian Stimberg, Dan Andrei Calian, and Timothy A Mann. Improving robustness using generated data. In *Advances in Neural Information Processing Systems*, volume 34, pages 4218–4233. Curran Associates, Inc., 2021. 9
- [23] Md Yousuf Harun, Jhair Gallardo, and Christopher Kanan. Controlling neural collapse enhances out-of-distribution detection and transfer learning. In *International Conference on Machine Learning*, 2025. 3
- [24] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016. 4
- [25] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. arXiv, 2016. 3, 6
- [26] Rui Huang, Andrew Geng, and Yixuan Li. On the importance of gradients for detecting distributional shifts in the wild. In *Advances in Neural Information Processing Systems*, volume 34, pages 677–689. Curran Associates, Inc., 2021. 3, 6
- [27] Conor Igoe, Youngseog Chung, Ian Char, and Jeff Schneider. How useful are gradients for ood detection really?, 2022. 3
- [28] Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, and Madian Khabisa. Llama guard: Llm-based input-output safeguard for human-ai conversations, 2023. 3
- [29] Debesh Jha, Pia H. Smedsrud, Michael A. Riegler, Pål Halvorsen, Thomas de Lange, Dag Johansen, and Håvard D. Johansen. Kvasir-seg: A segmented polyp dataset. In *Multi-Media Modeling*, pages 451–462, Cham, 2020. Springer International Publishing. 5
- [30] Adam Tauman Kalai, Ofir Nachum, Santosh S. Vempala, and Edwin Zhang. Why language models hallucinate, 2025. 1, 3
- [31] Elias Kempf, Simon Schrod, Max Argus, and Thomas Brox. When and how does clip enable domain and compositional generalization?, 2025. 1, 2
- [32] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization, 2014. 7
- [33] Jinsol Lee and Ghassan AlRegib. Gradients as a measure of uncertainty in neural networks. In *2020 IEEE International Conference on Image Processing (ICIP)*, pages 2416–2420, 2020. 3
- [34] Yucen Lily Li, Daohan Lu, Polina Kirichenko, Shikai Qiu, Tim G. J. Rudner, C. Bayan Bruss, and Andrew Gordon Wilson. Position: Supervised classifiers answer the wrong questions for out-of-distribution detection. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 81689–81713. PMLR, 2025. 3, 8
- [35] Shiyu Liang, Yixuan Li, and R. Srikant. Enhancing the reliability of out-of-distribution image detection in neural networks, 2017. 3
- [36] Weitang Liu, Xiaoyun Wang, John Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems*, volume 33, pages 21464–21475. Curran Associates, Inc., 2020. 3, 6

- [37] Xingming Long, Jie Zhang, Shiguang Shan, and Xilin Chen. Rethinking the evaluation of out-of-distribution detection: A sorites paradox. In *Advances in Neural Information Processing Systems*, 2024. 3
- [38] Lucy Ellen Lwakatare, Aiswarya Raj, Ivica Crnkovic, Jan Bosch, and Helena Holmström Olsson. Large-scale machine learning systems in real-world industrial settings: A review of challenges and solutions. *Information and Software Technology*, 127:106368, 2020. 1
- [39] Akshay Mehra, Bhavya Kaikhura, Pin-Yu Chen, and Jihun Hamm. Understanding the limits of unsupervised domain adaptation via data poisoning. *Advances in Neural Information Processing Systems*, 34:17347–17359, 2021. 2
- [40] John P. Miller, Rohan Taori, Aditi Raghunathan, Shiori Sagawa, Pang Wei Koh, Vaishaal Shankar, Percy Liang, Yair Carmon, and Ludwig Schmidt. Accuracy on the line: On the strong correlation between out-of-distribution and in-distribution generalization. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 7721–7735. PMLR, 2021. 3
- [41] Jose G. Moreno-Torres, Troy Raeder, Rocío Alaiz-Rodríguez, Nitesh V. Chawla, and Francisco Herrera. A unifying view on dataset shift in classification. *Pattern Recognition*, 45(1):521–530, 2012. 2
- [42] Eric Nalisnick, Akihiro Matsukawa, Yee Whye Teh, and Balaji Lakshminarayanan. Detecting out-of-distribution inputs to deep generative models using typicality, 2019. 3, 6
- [43] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019. 2
- [44] Stephan Rabanser, Stephan Günnemann, and Zachary Lipton. Failing loudly: An empirical study of methods for detecting dataset shift. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019. 3
- [45] Pietro Recalcati, Fabio Garcea, Luca Piano, Fabrizio Lamberti, and Lia Morra. Toward a realistic benchmark for out-of-distribution detection. In *2023 IEEE 10th International Conference on Data Science and Advanced Analytics (DSAA)*, pages 1–10, 2023. 3
- [46] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do cifar-10 classifiers generalize to cifar-10?, 2018. 2
- [47] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do imagenet classifiers generalize to imagenet?, 2019. 2
- [48] Ryne Roady, Tyler L. Hayes, Ronald Kemker, Ayesha Gonzales, and Christopher Kanan. Are open set classification methods effective on large-scale datasets? *PLOS ONE*, 15(9):e0238302, 2020. 3
- [49] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation, 2015. 4
- [50] Kate Saenko, Brian Kulis, Mario Fritz, and Trevor Darrell. Adapting visual category models to new domains. In *European Conference on Computer Vision (ECCV)*, Sept. 2010. 5
- [51] Zheyang Shen, Jiashuo Liu, Yue He, Xingxuan Zhang, Renzhe Xu, Han Yu, and Peng Cui. Towards out-of-distribution generalization: A survey, 2021. 1, 2
- [52] Juan Silva, Aymeric Histace, Olivier Romain, Xavier Dray, and Bertrand Granado. Toward embedded detection of polyps in wce images for early diagnosis of colorectal cancer. *International journal of computer assisted radiology and surgery*, 9(2):283–293, 2014. 5
- [53] Yiyun Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 20827–20840. PMLR, 17–23 Jul 2022. 6
- [54] Sunil Thulasidasan, Sushil Thapa, Sayera Dhaubadel, Gopinath Chennupati, Tanmoy Bhattacharya, and Jeff Bilmes. An effective baseline for robustness to distributional shift. In *2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA)*, pages 278–285, 2021. 1, 2
- [55] Birk Torpmann-Hagen, Pål Halvorsen, Michael A. Riegler, and Dag Johansen. Probabilistic runtime verification, evaluation and risk assessment of visual deep learning systems, 2025. 1, 3, 8
- [56] Birk Torpmann-Hagen, Michael Riegler, Pål Halvorsen, and Dag Johansen. Ood detectors are best used runtime verifiers, not semantic shift classifiers, 2025. 1, 2, 3
- [57] Birk Torpmann-Hagen, Michael Riegler, Pål Halvorsen, and Dag Johansen. Runtime verification for deep learning systems. In *Proceedings of the 20th International Conference on Evaluation of Novel Approaches to Software Engineering - Volume 1: ENASE*, pages 367–377. INSTICC, SciTePress, 2025. 1, 3
- [58] Birk Torpmann-Hagen, Michael A. Riegler, Pål Halvorsen, and Dag Johansen. Runtime verification for visual deep learning systems with loss prediction. *IEEE Access*, 13:48502–48519, 2025. 1, 3
- [59] V. Vapnik. *The Nature of Statistical Learning Theory*. Springer, New York, 1995. 2
- [60] Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. Open-set recognition: A good closed-set classifier is all you need? In *International Conference on Learning Representations*, 2022. 3
- [61] Hemanth Venkateswara, Jose Eusebio, Shayok Chakraborty, and Sethuraman Panchanathan. Deep hashing network for unsupervised domain adaptation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5018–5027, 2017. 5
- [62] Mei Wang and Weihong Deng. Deep visual domain adaptation: A survey. *Neurocomputing*, 312:135–153, 10 2018. 1, 2
- [63] Qizhou Wang, Yong Lin, Yongqiang Chen, Ludwig Schmidt, Bo Han, and Tong Zhang. A sober look at the robustness of clips to spurious features. In *Advances in Neural Informa-*

- tion Processing Systems, volume 37, pages 122484–122523. Curran Associates, Inc., 2024. 2
- [64] Julia Winkler, Christine Fink, Ferdinand Toberer, Alexander Enk, Teresa Kränke, Rainer Hofmann-Wellenhof, Luc Thomas, Aimilios Lallas, Andreas Blum, Wilhelm Stolz, and Holger Haenssle. Association between surgical skin markings in dermoscopic images and diagnostic performance of a deep learning convolutional neural network for melanoma recognition. *JAMA Dermatology*, 155, 08 2019. 1, 2
- [65] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M Alvarez, and Ping Luo. Segformer: Simple and efficient design for semantic segmentation with transformers. *Advances in neural information processing systems*, 34:12077–12090, 2021. 4
- [66] Jingkan Yang, Kaiyang Zhou, Yixuan Li, and Ziwei Liu. Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, 132(12):5635–5662, Dec 2024. 2, 3, 4
- [67] William Yang, Byron Zhang, and Olga Russakovsky. Imagenet-ood: Deciphering modern out-of-distribution detection algorithms. In *International Conference on Learning Representations*, 2024. 3, 8
- [68] Nanyang Ye, Kaican Li, Haoyue Bai, Runpeng Yu, Lanqing Hong, Fengwei Zhou, Zhenguo Li, and Jun Zhu. Ood-bench: Quantifying and understanding two dimensions of out-of-distribution generalization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7947–7958, June 2022. 2
- [69] Zhuowen Yuan, Zidi Xiong, Yi Zeng, Ning Yu, Ruoxi Jia, Dawn Song, and Bo Li. Rigorllm: Resilient guardrails for large language models against undesired content, 2024. 1, 3
- [70] Xingxuan Zhang, Yue He, Renzhe Xu, Han Yu, Zheyang Shen, and Peng Cui. Nico++: Towards better benchmarking for domain generalization, June 2023. 5

1242
1243
1244
1245
1246
1247
1248
1249
1250
1251
1252
1253
1254
1255
1256
1257
1258
1259
1260
1261
1262
1263
1264
1265
1266
1267
1268
1269
1270
1271
1272
1273
1274
1275
1276
1277
1278
1279
1280
1281
1282
1283
1284
1285
1286
1287
1288
1289
1290
1291
1292
1293
1294
1295